

Detection of Tropical Cyclone using Deep Learning Network

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Automatic weather forecasting is now achievable thanks to advancements in computer vision and satellite image technology. We have implemented a mechanism to forecast cyclones from satellite photos in order to lessen environmental damage and lower the risk to human life. A strong cyclone's eye inherits a lot of data regarding its behaviour and to locate it, we have built a Deep Learning object identification system, demonstrating that finding the cyclone's spinning centre is also a crucial task. A number of models that uses techniques like Deep learning, Machine learning have shown cyclone forecast with an accuracy of 86% to 95% and even greater, as well as cyclone eye detection accuracy above 87%. The outstanding findings of our experiment show that it can surpass traditional methods and previous works.

Keywords – CNN, Tropical cyclone, LSTM, Deep learning, Intensity, Forecast

1. Introduction

When there are variations in air pressure, a cyclone is a natural disaster that creates pressure gradient and makes the wind move and wind always blow from the areas of high pressure to low pressure areas, the movement is determined by low pressure and high pressure. This can also be referred as 'typhoon', 'hurricane', etc. and the center can rotate in either clockwise or counterclockwise direction.

Due to its destructive properties, including as extensive flooding, powerful winds, and intense downpours, it resulted in innumerable destruction all over the world. Understanding cloud patterns is crucial for making accurate predictions [1]. Early warnings for cyclone development have been provided by forecasters using satellite data, however much of the results were based on human judgment, which is inefficient in the modern world. To overcome some of the constraints in weather forecasting, researchers have used technology in machine learning and deep learning approaches, N. Haque et al. However, further research is necessary to create a system that is more trustworthy. Studies [2], [3] have reported problems with machine learning algorithms for forecasting the weather. Due to the availability of data, deep learning techniques have been applied in the remote sensing industry. It boasts astounding object detection and classification capabilities thanks to graphics processing units (GPUs). Strong GPUs are able to supply the computing power that such models need. Because of open-source deep learning software frameworks like TensorFlow and Keras, work has become simpler. We are motivated to work in this area by research on catastrophe detection using Deep Learning algorithms and classification of satellite images [4–6]. To recognise the cyclone and cyclone's eye in satellite pictures, we propose a Deep Learning-based system with many models. Our suggested methodology is mostly composed of three elements: The first three phases are image preparation, cyclone detection, and cyclone eye localization. With the help of numerous Deep Learning models, we were able to identify cyclones accurately. Finding a cyclone's centre is also crucial. Our Deep Learning-based approach outperformed multiple models in terms of recognition rate.

1.1. Motivation

The main motivation of the project is to contribute to meteorological department due to the recent deaths caused in cyclone and floods and to solve the problems and improve the accuracy of the existing systems.

1.2. Structure

The survey paper is structured with Section 2 describes the Existing System, Section 3 highlighting the literature survey that was related to the topic, Section 4 describes the Proposed system. Finally Section 5 concludes the study along with future scope (Section 6) and references (Section 7).

2. Existing System Model

Cyclone prediction is a rather well studied and surveyed area but still it remains a challenging task to make precise forecast on finding the intensity and path of the cyclone. Existing systems help the network to fit more tasks and handle paired data effectively but sometimes it states that the image is denser enough to move forward with the next image. In some cases, gradient vanishing leads to decrease in the accuracy. Limitations happens with the layers of neural organization and layers can be increased to achieve more accuracy. PMW channel has less frequency rate, VIS channel only provides information during daytime. Better channels can be used to improve the system.

3. Literature Survey

[1] Predicting tropical cyclone strength is challenging and complex. Track data is used for intensity forecasts. The following models were investigated Logistic Regression, Naive Bayes, Sequential Minimum Optimization, C4.5 Decision Tree, Multilayer Perceptron, Random Trees, and Random Forests. The latitude, longitude, centre pressure, pressure drop, and highest sustained wind speed all serve to identify the NIO. 2011-2020 best track data is used to measure intensity and compare machine learning classifiers. This study finds ML classifiers with 97.99% accuracy. Predicting tropical cyclone strength is challenging and complex. Track data is used for intensity forecasts. Logistic regression, Naive Bayes, Multilayer

perceptron, Random Trees, C4.5 decision tree, Sequential minimum optimization, and Random forests were studied. Latitude, longitude, central pressure, pressure drop, and highest sustained wind speed characterise the NIO. 2011-2020 best track data is used to measure intensity and compare machine learning classifiers. This study finds ML classifiers with 97.99% accuracy.

[2] Cyclone intensity is challenging to predict. Track data is used for forecasts. We looked at random forests, random trees, multilayer perceptrons, sequential minimum optimization, C4.5 decision trees, and logistic regression. NIO's latitude, longitude, central pressure, pressure drop, and greatest sustained wind speed. 2011-2020 best track data is used to compare machine learning classifiers. This research shows 97-98% accurate ML classifiers.

[3] For data in the Bay of Bengal and Arabian Sea, Ocean Observation Systems runs deep-sea moored buoy systems. OMNI measures subsurface salinity, temperature, current speed and direction. The buoys monitor cyclones and long-term climate. Innovative technical aspects were examined for buoy system performance in harsh sea situations. Average data return is 95%. Subsurface data reveals cyclone patterns and severity. Sensors monitor subsurface water.

[4] Industrial cyclone separators depend on accurate pressure drop predictions. Complex non-linear interactions between cyclone PDC and geometric factors need predictive modelling. Existing techniques involve theoretical/semi-empirical approaches that don't generalise well, and intelligent tactics for anticipating cyclone separator pressure drop haven't been examined. This study uses fuzzy modelling and an alternate method for training a multilayer neural network using the Extended Kalman Filter (EKF) (MLNN). Comparison of Lagrange dual SVM regression. After training, these models are designed using manual and grid search procedures. Fuzzy System (FS) prediction accuracy is 3.97×10^{-4} MSE and 99.70% R.

[5] Earthquakes, volcanic eruptions, extreme weather (lightning, hurricanes/cyclones, blizzards, tornadoes), and oceanic rogue waves make nonlinear dynamical systems more important. New machine learning. High-dimensional systems generate chaos. Machine learning predicts fibre ring cavity bursts. We demonstrated that spatiotemporal intermittency may change a stable periodic pattern with a well-defined frequency comb into chaos. Long-range dynamic correlation survives turbulence. Split. Chaos theory predicts repeating pictures, not entire systems. Pulses-precursors appear linked.

[6] Tropical cyclones imperil lives and property. Cyclone detection prevents disasters. This article discusses utilising a dual-branch CNN model based on GFS to identify cyclones, including tropical, extratropical, and subtropical cyclones. The model can learn geography, recognise essential

qualities, and make end-to-end predictions. The model accurately identifies tropical cyclones, extratropical cyclones, and cyclone disturbances over 90% of the time. It's easy and simple to use.

[7] Severe events such as earthquakes, volcanic eruptions, severe weather (lightning, hurricanes/cyclones, blizzards, tornadoes), and oceanic rogue waves make anticipating nonlinear dynamical systems crucial. Recent machine learning advancements are interesting. Size increases high-dimensional chaos and network nodes. We use supervised machine learning to forecast fibre ring cavity bursts. We recently demonstrated [3] that spatiotemporal intermittency may convert a stable periodic pattern with a well-defined frequency comb into a chaotic state. Figure 1a shows this. Despite turbulence, long-range dynamical correlation maintains. Subdivision follows. We use chaos theory to discover repeating pictures at the same burst place. We linked precursors-pulses via causality.

[8] We convert the GPU, many-core CPU, and nonhydrostatic atmospheric dynamical core of E3SM. Looking for 3-km and 1-km cloud resolutions. We utilise the C++ package Kokkos to express on-node parallelism in an architecture-independent approach. On IBM Power9 and Intel Knights Landing CPUs, our C++ solution is as least as fast as the original Fortran implementation, indicating that the code rework did not impact CPU efficiency. Our GPU-based solution achieves 0.97 Simulated Years Per Day on the complete Summit supercomputer. This is the highest resolution obtained by any global atmospheric dynamical core.

[9] About 90 tropical cyclones (TCs) develop annually across the tropical seas. Tracking and detecting TCs in a timely manner helps alert impacted areas. Remote sensing helps identify these storms over open waters distant from landmass. Here, we use deep learning to automatically recognise TCs in satellite pictures. In this study, We present a multistage deep learning system with an R-CNN detector, a wind speed filter, and an R-CNN classifier for TC detection (CNN). Bayesian techniques are used to optimise all pipeline hyperparameters. The suggested approach has good accuracy (86.55%), specificity (97.59%), and precision (97.10%) for test images.

[10] Tropical Cyclone (TC) monitoring has been improved since Dvorak's initial technique in the 1970s. One of the most often used operational TC intensity analysis methods is subjective and lacks visual and infrared pictures. This research utilises Sentinel-1 SAR data to detect tropical cyclone eye-centers and estimate their intensity. Gradient-based Class Activation Maps explain this learning approach. The suggested system effectively identifies TCs and can accurately locate their centres, competing with subjective and automated intensity evaluation approach.

References	Algorithm	Issues
[1]	Machine Learning Classifier	Tropical cyclone is stimulation is lacking.
[2]	Machine Learning	Only applicable for Tropical cyclone
[3]	Machine Learning	Analysing Data from history will not give accuracy because natural Disaster formed is always unique.
[4]	Back-Propagation Neural Network	Complex non-linear interactions can't be analysed accurately
[5]	Numerical weather prediction	Data Assimilation would be more complicated
[6]	Convolutional Neural Network (CNN) - Global Forecast System Analysis	Accuracy is only 75%
[7]	supervised machine learning	Predicting complex nonlinear dynamical systems will be complex
[8]	Kokkos-C++	Hardware requirement will be more
[9]	Deep Learning	Tracking and detecting TCs regularly would be more data consuming
[10]	Deep Learning	Over relayed on SAR Image

4. Proposed system

Utilizing satellite images from the IR and WV channels, a deep learning convolutional neural network model is utilised to investigate the feasibility of estimating tropical cyclone (TC) intensity. The LSTM technique is used to address gradient vanishing by disregarding irrelevant network information and analysing the features over time steps. The proposed system helps the network to handle paired data as well as connections among unpaired pictures and uses VGG-16 classifier for image classification. For binary classification, 500,000 non-TC data and 50,000 TCs and their predecessors are used to train the CNNs. The final result is sent as an alert message to the nearby authority in a precise format.

4.1 Modules description

4.1.1 Data acquisition

For the creation of a powerful ML model, the data collection process is essential.

The quality and quantity of your dataset has a direct effect on the decision-making process of the AI model. And these two factors influence the robustness, precision and performance of AI algorithms. As a consequence, data collection and structuring

often takes longer than data model training. We will gather data from Kaggle. To increase accuracy and robustness, the classification model's input dataset is preprocessed before being employed. The model takes picture input, which is followed by importing the necessary libraries. The raw images are checked for duplicacy and the data is split into 80% training data and 20% split data. The labelling of images comes next. The identification and labelling of particular details in an image is the focus of the data labelling technique known as "image labelling." Data labelling in computer vision entails adding tags to unprocessed data, such as images and videos. We will create labels for each image to classify the image as cyclone or non-cyclone.

4.1.2 IMAGE PREPROCESSING AND LABELLING

Data from actual scenarios always contain noise and missing values. This occurs due to manual mistakes, unexpected events, technical problems or various other obstacles. Incomplete and noisy data cannot be consumed by algorithms, as they are generally not designed to handle missing values, and noise causes disruption to the actual sample setup. Data preprocessing is a step in transforming raw data so that problems due to incompleteness, inconsistency and/or lack of proper representation of trends are resolved in order to obtain a data set in a comprehensible format. The images in the dataset underwent the following various operations during pre-processing:-

→Resizing of images

Resizing all images to same size helps to reduce computation power.

The cropped photos are downsized to the input size of 224x224 pixels, which is needed for our pretrained network.

→Convert Color Channel

The process of grayscaling involves changing an image from one colour space to another, such as RGB, CMYK, HSV, and so on. It can be completely black or completely white.

The OpenCV function imread() reads the image file, and the order of the colours is BGR (blue, green, red). On the other hand, Pillow assumes that the order of the colours is RGB (red, green, blue). Therefore, BGR and RGB must be converted to use both the Pillow function and the OpenCV function. To apply the Open CV function cvtColor(), begin by importing OpenCV and use the imread() function to read the original image before using the cv2.cvtColor() method to convert it to grayscale.

After the script has ended, users can always close or destroy all windows using the destroyAllWindows() function.

→Zooming

Image zooming is the process of enlarging an existing image. Thermal photographs that have been zoomed in can be utilised as enhancement because they show what happens when photos are taken up close.

→Contrast enhancement

Enhancing the contrast in an image allows it to have more contrast and more intensities overall than it would without.

Since the same photograph can have stronger contrast on a sunny or bright day, this was used to enhance images and this method is used in the training set to include a wide variety of potential outcomes.

→Salt and pepper noise

Salt and pepper noise is applied to an image by randomly changing some pixel intensities to 1 and others to 0. Salt and pepper noise may appear in photos taken on dusty days or when the camera is covered in dust.

4.1.3DEEP LEARNING CLASSIFICATION

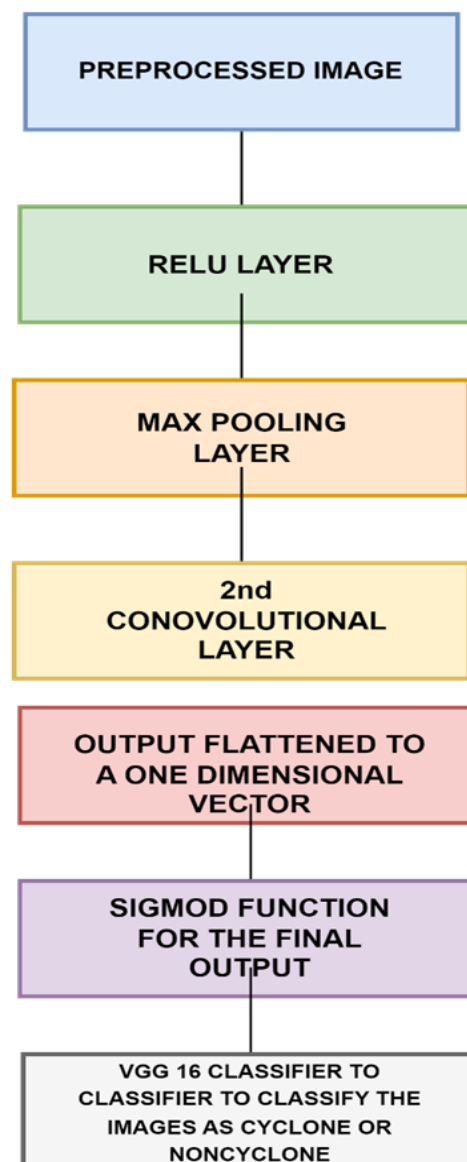
For building the model, CNN, LSTM and VGG-16 is chosen as the algorithms. As deep learning plays a huge role in object detection, and classification tasks, it is implemented. First, the input image is preprocessed and then taken to the ReLU layer of the CNN model. Once the input image is processed by the CNN model, the VGG-16 classifier is used to classify the image as cyclone or non-cyclone. The LSTM model predicts when the cyclone is supposed to be in the final form and also helps to detect the eye of the cyclone.

Convolutional neural network architecture VGG16 is used. It has a reasonably straightforward architecture and excels in image classification. The feature extractor is the first half of the network (black and red). An image is fed to it. It is in charge of obtaining pertinent information from that image. Each layer that receives data uses less memory to hold image data than the one before it. The second half of the network, composed of the colors blue and gold, is in charge of using some latent representation to carry out a task. The task for VGG16 is frequently classification. This could entail classifying an image by printing "cat" or "dog." This portion of the network learns how to interpret the latent representations mathematically. This point on, it can categorize, draw a bounding box, etc.

Photos will be sent through the convolutional layers of VGG16, which will provide a Feature Stack containing the recognized visual features and training model is downloaded. Setting up Learning rate and optimizer is essential and it should contain the following: -

- Binary cross-entropy, a logarithmic loss function, is used in the model as the loss function.
- The Adam optimizer has been used for optimization with a variety of parameters, including learning rate, decay, etc.

Thus, the output is an image predicted as cyclone or non-cyclone and will set up an alert signal as a message to inform the authorities.



Fig,1-Steps in classification of an image

5. Conclusion

Automated cyclone prediction is continually in demand since cyclones are one of the most destructive natural catastrophes. With the aid of satellite photos and deep learning methods, we have proposed a solution to this significant problem. We have developed a two-step method to accurately pinpoint cyclones and their center cloud pattern. Although it was challenging to locate a benchmark dataset for our work, the technique we used produced outstanding results. However, in order to broaden the scope of our future study, we would like to utilize more data, such as surface temperature and air pressure, in order to forecast, track, and estimate a more exact center position for cyclones. Overall, the study demonstrates a definite advantage for using the recommended Deep Learning models for cyclone prediction.

6. Future Scope

Even though there are many advanced technologies developed for the detection of cyclone, the path of the tropical cyclone is still difficult to find with a higher accuracy rate because of the lack of accurate information about the external factors around the cyclone. These external factors include humidity, sea level rise, temperature, wind speed, wind direction. The combined information of these factors would pave a greater way in the future to improve the prediction rate.

7. References

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